

Quiz #2 on Algebraic Specification

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1 Arrays

We want an algebraic specification of arrays. An array is a list whose size is fixed at the construction time and whose elements are items of the same type. There is a special item which is used as a default element when creating a new array (i.e. a freshly created array contains these default elements). Then the user can change an element to another value by specifying the integer index (i.e. position) in the array and the new value.

Let us call ITEM the specification of some items whose signature is as follows.

- **Defined types** The type of the items is noted t
- **Constructors**
DEFAULT : t
The term DEFAULT is a distinguished item, whose interpretation, i.e. meaning, is left to the user of this specification.

Let us call now ARRAY(ITEM) the specification of the arrays over items of type ITEM. t . The signature is as follows.

- **Defined types** The type of the arrays is noted t .
- **Constructors**
 - EMPTY : t
The term EMPTY represents the array which contains no element.
 - CREATE : $\text{int} \times \text{int} \rightarrow t$
The term CREATE(x, y) denotes an array whose elements are indexed from x to y , if $x < y$, or from y to x otherwise. The type int denotes the set of positive integers. Integers x and y are called *bounds*. If $x < y$ then x is the *lower bound* and y is the *upper bound*.
For example, CREATE(3, 5) is an array whose elements are indexed by integers 3, 4 and 5. It contains three elements. These elements are all equal to ITEM.DEFAULT. Also CREATE(5, 3) is the same

as $\text{CREATE}(3, 5)$. Therefore, in both cases, the lower bound is 3 and the upper bound is 5.

– $\text{SET} : \mathbf{t} \times \text{int} \times \text{ITEM.t} \rightarrow \mathbf{t}$

The term $\text{SET}(a, n, e)$ represents an array equal to array a except that the element at index n is e . If n is out of bounds, i.e. if n is not between the bounds of a , the result of SET is unspecified. Same if a is empty.

• Functions

– $\text{LOWER} : \mathbf{t} \rightarrow \text{int}$

The call $\text{LOWER}(a)$ is the lower bound of array a , i.e. the smallest valid index. If a is empty, the result is unspecified.

– $\text{UPPER} : \mathbf{t} \rightarrow \text{int}$

The call $\text{UPPER}(a)$ is the upper bound of array a , i.e. the greatest valid index. If a is empty, the result is unspecified.

– $\text{GET} : \mathbf{t} \times \text{int} \rightarrow \text{ITEM.t}$

The call $\text{GET}(a, n)$ denotes the element in array a at index n . If n is out of bounds or if a is empty, the result of GET is unspecified.

– $\text{NTH} : \mathbf{t} \times \text{int} \rightarrow \text{ITEM.t}$

The call $\text{NTH}(a, n)$ denotes the n th element in array a . If a is empty or n is out of bounds, the call is unspecified.

– $\text{MEM} : \mathbf{t} \times \text{ITEM.t} \rightarrow \text{bool}$

The call $\text{MEM}(a, e)$ is TRUE if e is an element of array a and FALSE otherwise.

– $\text{INV} : \mathbf{t} \rightarrow \mathbf{t}$

The call $\text{INV}(a)$ denotes the array whose elements are in the reverse order as find in a . For example, let a be the array graphically represented as

e_1	e_2	e_3	e_4
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, then $\text{INV}(a)$ is represented as

e_4	e_3	e_2	e_1
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– $\text{UNDO} : \mathbf{t} \times \text{int} \rightarrow \mathbf{t}$

The call $\text{UNDO}(a, n)$ represents the array a without the last modification (using SET) at index n . If a is empty or has not been modified at index n , the call equals a .

Question. Complete this signature with equations defining the functions NTH , MEM , INV and UNDO (remember you can use LOWER , UPPER or GET if you need).